

qicert: Certified Quantum-Inspired Compression of Vision–Language–Action Models with Exact Bounds, Provable Pruning, and Formal Safety Tails

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Code: <https://github.com/ZakLr/qicert> Reproduces every table via `qicert.bench.all`

Abstract—Vision–Language–Action Models (VLAMs) at 7B–10B parameters are too large for embedded vehicle and robot controllers and too expensive to certify. We present *qicert*, a backbone-agnostic pipeline that compresses VLAMs into quantized tensor-train (QTT) networks before the heaviest training, compiles cross-modal interaction blocks into exactly-evaluable commuting-Pauli families, and certifies what remains with a three-layer certificate stack: exact algebraic bounds (Lipschitz products of QTT cores; per-family pruning errors), local formal proofs (SOS boxes; Lyapunov-margin survival curves), and statistical tail bounds on signal-temporal-logic robustness estimated by iterative quantum amplitude estimation raced against the strongest honest classical estimator (RESTART+GEV) and closed by scenario-optimization and conformal calibration. A syndrome-shadow monitor guards the action stream with theorem-budgeted false-alarm rates. The INT8 baseline can match our accuracy at matched compression; it cannot compute our certificates, which are exact functions of the compressed cores.

Index Terms—tensor networks, quantized tensor train, Lipschitz certificates, Pauli grouping, Clifford diagonalization, quantum amplitude estimation, signal temporal logic, classical shadows, VLAM, autonomous driving, robotics, formal safety

I. INTRODUCTION

VLAMs fuse camera, LiDAR, radar, and language inputs into planning and control outputs. Reference models such as Alpamayo-R1 [1] and π_0 [2] demonstrate the category’s potential, but their scale makes them prohibitively expensive to train and nearly impossible to deploy on embedded hardware within the 100 ms real-time budget while satisfying functional-safety certification (ISO 26262; IEC 62061 / ISO 10218).

The challenge asks for a reproducible quantum-enhanced or quantum-inspired component with a clear advantage over a classical baseline across four bottlenecks: model footprint, attention complexity, RL alignment cost, and control-safety guarantees. Quantum-inspired tensor networks have demonstrated strong LLM compression [3], and provably efficient quantum algorithms promise gains in the high-dimensional linear algebra central to VLAM training [4]. Our answer is a *system*: one pipeline in which five quantum-inspired primitives — tensor networks, Pauli observables, amplitude encoding, syndrome measurement, and shadow tomography — are the type system of every stage (train $\rightarrow$ compress $\rightarrow$ compile $\rightarrow$ certify $\rightarrow$ monitor). The contributions are:

- 1) **Exact-certificate compression**: QTT cores double as the objects whose spectral norms yield an exact global Lipschitz constant (Sec. III), giving a certified-safe-set column that no quantization-based baseline can produce at any compression ratio.
- 2) **Compilation-as-identity**: cross-modal interaction tensors are grouped into commuting Pauli families, simultaneously diagonalized by single Cliffords, and evaluated exactly; dropping a family costs an exact, printed error $\|\sum_{a \in F_j} c_a\|_1$ (Sec. IV).
- 3) **Quadratic query savings on the safety tail**: rare-event failure probability is estimated by iterative quantum amplitude estimation (IQAE) with Bayesian credible intervals, raced against RESTART+GEV and closed by Campi–Garatti scenario-optimization and conformal prediction (Sec. V).
- 4) **A runnable, self-reproducing artifact**: *qicert* (“certified quantum-inspired compression”) reproduces every table from a clean environment and emits its own hardware pathway (qubits, depth, shots).

All quantum components are simulated per challenge constraints (Sec. 5.3); resource assumptions are stated per algorithm. Results below are reported as pre-registered targets with measured values pending (Sec. IX).

II. SYSTEM ARCHITECTURE

The pipeline is identical across both tracks; only the slice specification differs.

Training time (offline, Kaggle 2×T4): Stage 0 baseline fine-tune (QLoRA/LoRA) of the backbone (AD: LLaVA-1.5-7B with Alpamayo-adjacent action head, justified per Sec. 5.4; robotics: MiniVLA-class head). **Stage 1** compress into {TT, QTT} via TT-cross (query-based, no global SVD); the safety-budgeted bond allocator sets per-layer ranks. **Stage 2** ALS/DMRG-native fine-tuning on the compressed manifold — the dense matrix is never formed. **Stage 3** compile cross-modal interaction blocks into commuting-Pauli families. **Stage 4** certify (three layers, Sec. III). **Stage 5** package into *qicert*; the bench suite reproduces every table.

Runtime (deployed, ≤ 100 ms profile): sensor + instruction encoders $\rightarrow$ compiled interaction kernels (diagonal, $O(N \cdot k)$ -style) $\rightarrow$ QTT-compressed action head with certified margins

TABLE I
THE SEVEN FUSION JOINTS. EACH COMPONENT CONSUMES ANOTHER’S OUTPUT, SO ABLATIONS ISOLATE CLEANLY AND THE MARGINAL COST OF EVERY EXTRA CLAIM IS NEAR ZERO.

Joint	What it buys
1	QTT cores = certificate inputs (zero extra compute)
2	Compiled families = monitor’s measurement basis (zero extra design)
3	Certificate margins = the STL predicates IQAE estimates (compositional)
4	Sec. 5.2 seeded scenarios = the amplitude-estimation oracle
5	Worst-case long-tail margin = bond-allocation objective (tail designs, model)
6	Compiled qubit/depth/shots table = the hardware pathway
7	qicert carries everything

Stage 0: baseline fine-tune (QLoRA/LoRA)
↓ Stage 1: {TT,QTT} TT-cross compression + bond allocator
↓ Stage 2: ALS/DMRG-native fine-tuning (dense matrix never formed)
↓ Stage 3: compile commuting-Pauli families (exact, prunable)
↓ Stage 4: certify — L1 exact Lipschitz / L2 SOS + Lyapunov / L3 IQAE tails
↓ Stage 5: package qicert; bench suite reproduces every table

Fig. 1. Training-time pipeline (Sec. II); the runtime view adds the syndrome-shadow monitor (Sec. VI).

→ syndrome-shadow monitor (PDU hygiene + shadow statistics) → safe output or certified fallback action.

Fusion joints (why this is a system, not a portfolio):

Unifying principle. A VLAM that will be certified and deployed on embedded hardware is, at every stage, a computation on *structured tensors* — weights, interaction tensors, belief and robustness distributions. Quantum information is the discipline that made structured tensors a first-class programming model: tensor networks, Pauli observables, amplitude encoding, syndrome measurement, shadow tomography. We build the entire train → compress → compile → certify → monitor pipeline from those five primitives. Quantum-inspiration is the type system of the whole submission, not a label on one component.

III. THE CERTIFICATE STACK

Three layers answer three questions and are never confused.

A. Layer 1: Exact algebraic bounds

For a TT/QTT layer with cores $G_1 \dots G_d$, the induced map is multilinear with exact global Lipschitz constant

$$L(W) = \prod_{i=1}^d \|G_i\|_2, \quad (1)$$

computed in $O(\sum_i r_i^2 n_i)$ per layer (each core’s spectral norm by power iteration on the small core). Chaining through the network with 1-Lipschitz activations (ReLU et al.),

$$L(\tilde{F}) = \prod_{\text{layers } l} \prod_i \|G_i^{(l)}\|_2. \quad (2)$$

[Measured in N3/N5: certified-safe-set % vs compression ratio]

qicert-QTT: monotone decay, bond-spectrum predictor band
INT8 / SVD: not computable (flat 0% line, stated honestly)

Fig. 2. The asymmetry plot (pre-registered): the certificate column that quantization baselines structurally cannot produce.

With the input box B from the slice spec this yields the *exact safe-set certificate*

$$\text{Safe}(B) = \{x \in B : \|\tilde{F}(x) - \tilde{F}(x_{\text{ref}})\| \leq L(\tilde{F}) \text{diam}(B) < \text{margin}\}. \quad (3)$$

Theorem 1 (Lipschitz product, layer 1). *For a TT layer with cores $G_1 \dots G_d$, Eq. (1) holds exactly, with no sampling and no input-distribution assumption. The property is preserved for QTT (log-mode) reshapes because it depends only on the contraction, not on mode sizes.*

The load-bearing comparison: INT8 quantization error is data-dependent (clip ranges, per-channel scales) and unfactorizable — no exact global constant exists. SVD truncation gives no certificate object to compose. Our compressed cores *are* the certificate.

B. Layer 2: Local formal proofs

(a) SOS-certified boxes. Composition polynomials of TT-structured networks admit moment-SOS hierarchies with tractable SDPs [8]. For each sampled long-tail box B_k we certify

$$\forall x \in B_k : \text{margin}(x) \geq m_k \geq 0, \quad (4)$$

proven by an exported SOS certificate (machine-checkable Gram matrix). Layer 1 is global-but-loose; Layer 2a is local-but-tight; disagreement between them is a surfaced bug, not a hidden number.

(b) Lyapunov-margin degradation curve + predictor. The old binary “certificate survives” claim is upgraded to a continuous object. At each compression ratio r ,

$$\text{survival}(r) = \frac{\#\{\text{trajectories where } V\text{-decrease holds}\}}{\#\{\text{trajectories}\}}, \quad (5)$$

with a regression from (bond spectra, entropy signals, layer sensitivities) to $\text{survival}(r)$. The submission prints the *predicted* largest safe ratio r^* *before* measuring it, then reports $|r^* - r_{\text{measured}}|$.

C. Layer 3: Statistical tail bounds on formal robustness

Safety requirements are Signal-Temporal-Logic (STL) formulas with quantitative robustness ρ (Fainekos–Pappas semantics; $\rho > 0$ = satisfied with margin) [13]. The challenge metric “% episodes in a safe equilibrium” is the special case $\Pr[\rho \geq 0]$; we report the full ρ distribution and its tails.

TABLE II

COMPILER PASSES. PASS 4 IS AN IDENTITY, NOT AN APPROXIMATION.

Pass	Operation
1	Extract interaction table $\{(c_a, P_a)\}$
2	Group into mutually commuting families $\{F_1, \dots, F_m\}$ (graph coloring)
3	Per family: single Clifford C_j with $C_j^\dagger P C_j \in \{I, Z\}^{\otimes k}$
4	Rewrite exactly $T = \sum_j C_j^\dagger D_j C_j$, D_j diagonal
5	Emit hardware table: qubits = $\lceil \log_2(\text{register}) \rceil$, depth = m , shots $\leq f(\ c\ _1, \epsilon)$
6	Emit pruning certificates: dropping F_j costs exactly $\left\ \sum_{a \in F_j} c_a \right\ _1$

[Measured in N7: predicted $\left\| \sum_{a \in F_j} c_a \right\|_1$ vs measured Δ]
 Target: correlation ≥ 0.95 (E-comp-2); per-family leave-one-out

Fig. 3. Predicted vs measured pruning error — the ablation column no other submission can print.

IV. THE COMMUTING-PAULI INTERACTION COMPILER

Each cross-modal pairwise interaction term is a Pauli string $P_a = \sigma_{a_1} \otimes \dots \otimes \sigma_{a_k}$ with coefficient c_a (learned or inherited); the interaction tensor is $T = \sum_a c_a P_a$. Grouping into commuting families and per-family Clifford diagonalization are standard VQE measurement-reduction machinery [11], [12]; our contribution is their use as an exact, prunable compiler for cross-modal interaction tensors. The compile pass:

Exactness (E-comp-1): $\|T_{\text{exact}} - T_{\text{compiled}}\| \leq 10^{-6}$ relative (numerics/coefficient-estimation only). Runtime cost per family scales with the diagonal evaluation, not pairwise $O(N^2)$ enumeration — the attention bottleneck answered with an identity plus an $O(N \cdot k)$ -style diagonal kernel. Pruning is compositional with Eq. (3): the certified safe set survives a *documented* pruning budget.

Why this survives the QASA objection. QASA showed a capacity-matched classical low-rank bottleneck matches a PQC’s error metrics [9]. Our claim is not “the encoding is powerful” but “the compiler is exact, certifiable, and executable”: generic feature crosses have no commuting-group structure, and Performer/FAVOR+ [10] approximates the softmax kernel with distributional error and no notion of a family — it cannot prune with a printed certificate, and it emits no hardware table.

V. SAFETY EVALUATION ENGINE

A. Oracle

The Sec. 5.2 seeded-scenario requirement makes the simulator a reversible deterministic function of the seed:

$$O : \text{seed} \mapsto \text{scenario} \mapsto \text{rollout}(\pi) \mapsto \rho \mapsto [\rho < 0]. \quad (6)$$

B. The three-arm race

Amplitude estimation estimates p to absolute error ϵ with $O(1/\epsilon)$ oracle queries versus Monte-Carlo’s $O(1/\epsilon^2)$ — quadratic query savings on the rarest scored quantity. The claim is framed as query-complexity against a counting problem, with the oracle cost model printed per benchmark. If Q

TABLE III

THREE-ARM SAFETY ESTIMATION (PRE-REGISTERED STRUCTURE; COMPLEXITY AT $p = 10^{-5}$).

Arm	Method	Output	Query cost
		raw table	
C-strong	IQAE, Bayes adaptation [14], [15]	raw table	$\sim 10^2$ – 10^3 oracle queries
C-naive	RESTAR/Cliffordable GEV tail fit [18]	Cliffordable kernels	10^3 – 10^4 rollouts (measured)
	Monte-Carlo counting	Sec. 4.2 pathway	$\sim 10^6$ + (cost shown, not)
		ablation knob	

loses to C-strong at our budgets, we print that (pre-registered R4); Layers 1–2 stand independently. Adjacent QAE applications include structural reliability [15], flow-based failure sampling [16], and MPPI control [17]; our in-domain delta is VLAM/AD safety certification fused with exact structural certificates.

C. Closure

Scenario optimization (Campi–Garatti): with N scenarios and solution complexity d , $\Pr[\text{violation} > \epsilon] \leq \beta$ whenever $N \geq N(\epsilon, \beta, d)$ — printed per claim as an (N, ϵ, β) triple [19]. **Conformal calibration**: all intervals receive split-conformal coverage correction on a held-out slice (never the training slice; leakage rule R12). **Falsification loop**: failures found by RESTART/IQAE feed back as training-slice augmentations, new long-tail slice entries, and monitor training positives; the report’s figure shows tail mass shrinking across rounds.

VI. RUNTIME MONITOR

A two-layer guard on the action stream with theorem-budgeted guarantees.

Layer 1 — PDU hygiene (zero extra FLOPs). The compiled commuting-family table (Sec. IV, pass 3/6) already computes a syndrome per inference step. If the syndrome changes between consecutive steps, the packet is gray-listed: this catches bit-flip corruption of the Pauli register — sensor dropouts, camera blinding, CAN-bus corruption — the exact failure modes of the challenge’s synthetic perturbation suite.

Layer 2 — Classical-shadow statistics. With M random projection directions $v_1 \dots v_M$ (Huang–Kueng–Preskill shadows [20]), median-of-means concentration over the scalars $s_i(x) = \langle G_i, x \rangle$ gives a statistically correct false-alarm budget at memory $O(M \cdot s)$ and a few matvecs of latency — continuous OOD shift that a parity bit cannot see. A split-conformal gate sets the user-visible false-alarm rate (e.g. $\leq 1\%$). Gray-list fallback: the largest-radius action from the Layer-1 certified safe set — the monitor never aborts into the void.

VII. TRAINING ENGINE

QTT reshape [6]: $m = \prod m_i$, $n = \prod n_j$ (binary or small-factor modes, bit-ordering per layer type; N2’ resolves the ordering). **TT-cross** [5], [7]: cores $G_1 \dots G_d$ are built from $O(\sum m_i n_i r_i^2)$ maxvol queries of W ’s entries, never forming W — SVD-free and VRAM-safe for 7B-scale layers.

TABLE IV
EXPERIMENT MATRIX (PRE-REGISTERED; GPU-H ARE QUOTA-HOURS).

ID	Experiment
N1	AD backbone fine-tune + INT8 reference
N2	{TT,QTT}×TT-cross sweep, 6 bond plans × 2 backbones
N2'	Bit-ordering sensitivity (3 orders, 1 layer, 1 seed)
N3	Layer-1 certificate table (Lipschitz products per Pareto point)
N4	Layer-2a SOS boxes (3 points × 2 tracks × slices)
N5	Layer-2b Lyapunov curve + bond-spectrum predictor
N6	Layer-3 safety suite (IQAE vs RESTART+GEV vs MC)
N7	Compiler exactness + prune certificates + latency
N8	(optional) PQC RL toy head + noise sweep
N10	Syndrome-shadow monitor demo
N11	ALS/DMRG-native fine-tune, both backbones
N12	Safety-critical last-pass on falsified slice
N13	Cross-track mirror (N2/N3/N5 on second backbone)
N14	Clean-env end-to-end bench audit

ALS/DMRG-native fine-tuning: each core is optimized directly (regularized least squares with exact multilinear gradients; AMEn enrichment modulates bond dimension where the allocator’s safety signal flags it). Memory scales with core size, not layer size. Training passes: P1 baseline fine-tune; P2 compress → ALS fine-tune (the deliverable); P3 safety-critical last-pass on the RESTART/IQAE-falsified failure slice only (tail-shrinkage evidence, ~hundreds of hard examples).

Safety-budgeted bond allocator: per layer,

$$r^* = \arg \min_r \max_{\text{slices}} (\text{certified long-tail margin at } r), \quad (7)$$

using (1) layer sensitivity to the certified-margin computation, (2) bond-spectrum entropy, (3) monitor hygiene bits. The long-tail slice is a design selector, not a report line.

VIII. EXPERIMENTAL SETUP

Tracks and baselines (Sec. 5.4): AD — LLaVA-1.5-7B (Alpamayo-adjacent head) on nuScenes [21]/CARLA-style slices, vs. INT8 bitsandbytes; robotics — MiniVLA-class backbone (MIT code, HF checkpoints; Open X-Embodiment lineage [22]) on LIBERO-style slices, vs. INT8. Strong extra baselines: SVD-at-matched-ratio, tuned Performer at matched capacity, RESTART+GEV (not naive MC). **Hardware:** Kaggle 2×T4 (30 GPU-h/wk, 120 h ceiling), local RTX 5060 for small runs; all budgets are quota-hours, silicon-hours logged separately. **Discipline:** mean ± std over ≥3 seeds; every report table has a ledger row (research/experiments.md); every experiment has a pre-registered kill criterion.

Budget rollout: 87 GPU-h core, 33 h headroom vs. the 120 h ceiling; weeks 1–6 schedule in 08-experiments.md. Week 1: N1+N2'+N3 (the cheapest exposure of the two load-bearing facts: does the baseline behave, and does QTT compress with certificates attached?). The optional N8 PQC demo uses a quantum natural policy gradient head [23] for illustration only; it is cut first under quota pressure (R10).

IX. RESULTS FRAMEWORK

Per challenge Sec. 5.5, all values are mean ± std over ≥3 runs. Tables below state the pre-registered targets; measured

TABLE V
MAIN RESULTS TABLE (STRUCTURE; † = PRE-REGISTERED TARGET).

Roller	GPU-h	qicert-QTT	INT8	SV
Accuracy @ 28 (drop vs dense)	28	≤5%†	measured	meas
Accuracy @ 42 (drop vs dense)	42	target ≤5%†	measured	meas
Certified-safe-set % @ 2×	2	>90%†	— (not computable)	—
Certified-safe-set % @ 4×	4	target >50%†	—	—
Latency (ms, stated profile)	3	≤100	measured	meas
Accuracy @ 6	6			
C	8			
B	6			
accuracy	4			
monitor	3			
engine	8			
E-comp	3			
A	12			
exactness	4			
prune-certificate corr.				
3 latency				
4 hardware self-consistency				

Arm	Output
Q (IQAE)	credible interval on $p = \Pr[\rho < 0]$
C-strong (RESTART+GEV)	profile-likelihood CI on p
C-naive (MC)	cost shown; Clopper–Pearson CI
Scenario-opt	(N, ε, β) triple
Conformal	coverage on held-out slice

values land from the ledger during weeks 1–6 and replace the placeholders. No number in this report is fabricated; every number in the final report will be regenerable via `qicert.bench.<table>`.

X. ABLATION AND QUANTUM JUSTIFICATION

Per challenge Sec. 5.5, the QI-isolating ablation is mandatory and is designed to be the strongest section of the report.

A1 (compression×certificate): at matched ratio, INT8 matches accuracy but cannot report the certificate column — the ablation shows the asymmetry directly (the number that cannot be computed is the win).

A2 (compiler vs encoding): per-family leave-one-out with predicted error $\|\Sigma_{a \in F_j} c_a\|_1$ vs measured Δ ; and compiler-vs- encoding: the exact identity vs a bare feature cross (shows the family structure is what buys exactness, not the encoding label). Tuned Performer at matched capacity is the classical ceiling; winning it is upside, losing it does not kill the claim (the primary axis is exactness + certificates + hardware table).

A3 (IQAE vs strong classical): the three-arm race at matched query budget; if Q loses, the report prints that (R4) and the claim rests on Layers 1–2 and the strong classical arm.

A4 (monitor): syndrome-shadow vs autoencoder baseline: correction rate at matched overhead; false-alarm rate vs the shadow theorem bound (≤3× or demoted, R7).

A5 (training engine): ALS-in-compressed vs dense-then-compress at matched ratio on the safety slice (≤2% worse or demoted, R6).

XI. LIMITATIONS AND RISKS

Pre-registered kill/demote criteria (full register in 09-risks-kills.md): R1 compression accuracy $>5\%$ drop at $2\times \rightarrow$ CompactifAI-style softer claim + certificate story; R2 Layer-1 bound too loose ($<50\%$ certified safe set) $\rightarrow$ SOS becomes primary; R3 SOS SDP intractable $\rightarrow$ Lyapunov curve primary, SOS reported as “attempted, too slow”; R4 IQAE loses to RESTART $\rightarrow$ RESTART stands, IQAE = future work; R5 family count too large $\rightarrow$ top- k by $\|c\|_1$ with certificate-bound; R6 ALS-in-compressed $>2\%$ worse $\rightarrow$ demote to post-processing; R7 shadow FPR $>3\times$ theorem $\rightarrow$ PDU hygiene only; R8 allocator tradeoff $\rightarrow$ reported honestly; R9 cross-track non-comparable $\rightarrow$ pipeline claim only; R10 quota overrun $\rightarrow$ N8 cut first, then N13 reduced; R11 clean-env failure $\rightarrow$ report held; R12 conformal leakage $\rightarrow$ claim degraded until re-run.

Three things that cannot kill the core claim: (1) the exact Lipschitz-product certificate — a theorem, not a benchmark; (2) compilation-as-identity + pruning certificate — a mechanism, not a number; (3) the three-layer safety methodology — a formal spec plus raced estimators plus conformal closure, not a score. The final table is built from whatever layers survive their criteria; the package always runs and reports something certified.

Certification framing: the STL specs map to ISO 26262 / IEC 62061 / ISO 10218 artifacts (hazard $\rightarrow$ safety goal $\rightarrow$ STL formula $\rightarrow$ robustness predicate $\rightarrow$ certified tail probability), and the certificate stack provides the evidence trail an assessor expects.

XII. RESOURCE DECLARATION

Hardware: Kaggle $2\times$ NVIDIA T4 16 GB (primary), local RTX 5060 (secondary). **Total:** 87 GPU-h core (quota-hours) + headroom; silicon hours logged separately in the ledger. **Simulation:** CUDA-Q (pinned version), PennyLane, torch — versions in `environment.yml`. **Shot/gate budgets:** per benchmark, stated in the compiler’s pass-5 table. **Energy:** measured by the qicert profiler (kWh + CO₂e at declared grid intensity). Reproducibility evidence: `qicert.bench.all` from a fresh virtualenv, verified by N14.

XIII. CONCLUSION

We submit objects the baselines structurally cannot produce: exact certificates that are functions of the compressed cores, an exact compilation identity with per-family provable pruning, quadratic query savings on the rarest scored quantity, and a package that reproduces every table from a clean environment. All claims are either theorems, structural properties, or pre-registered measurements with kill criteria — the honesty contract of Sec. 5.5 operationalized. The full argument, theory, and risk register are maintained in the submission folder of the qicert repository.

REFERENCES

[1] NVIDIA and Y. Wang et al., “Alpamayo-R1: Bridging Reasoning and Action Prediction for Generalizable Autonomous Driving in the Long Tail,” arXiv:2511.00088, 2025.

[2] K. Black et al., “ π_0 : A Vision-Language-Action Flow Model for General Robot Control,” arXiv:2410.24164, 2024.

[3] A. Tomut et al., “CompactifAI: Extreme Compression of Large Language Models using Quantum-Inspired Tensor Networks,” arXiv:2401.14109, 2024.

[4] J. Liu et al., “Towards Provably Efficient Quantum Algorithms for Large-Scale Machine-Learning Models,” Nature Communications 15(1):434, 2024.

[5] I. V. Oseledets, “Tensor-Train Decomposition,” SIAM J. Sci. Comput. 33(5):2295–2317, 2011.

[6] I. V. Oseledets, “Approximation of $2^d \times 2^d$ matrices using tensor products,” arXiv:1007.1170, 2010.

[7] M. Bousse et al., “TTOpt: A MATLAB package for Tensor-Train optimization,” arXiv:2205.00293, 2022.

[8] (SOS hierarchies on TT-structured polynomials), arXiv:2604.17563, 2026.

[9] S. Chen and Y. Kuo, “Quantum-inspired attention architectures,” Quantum Science and Technology 11:035051, 2026.

[10] K. Choromanski et al., “Rethinking Attention with Performers,” arXiv:2009.14794, 2020.

[11] A. Reggio et al., “Pauli grouping for measurement reduction,” arXiv:2305.11847, 2023.

[12] (k-NoClID: grouped Clifford diagonalization), arXiv:2408.11898, 2024.

[13] G. E. Fainekos and G. J. Pappas, “Robustness of temporal logic specifications,” in Proc. Formal Methods and Models for Co-Design (MEMOCODE), 2009.

[14] D. Grinko, J. Gacon, C. Zoufal, and S. Woerner, “Iterative Quantum Amplitude Estimation,” npj Quantum Information 7:52, 2021; arXiv:1912.05546.

[15] (Bayesian quantum amplitude estimation for structural failure probabilities), arXiv:2607.18996, 2026.

[16] (Quantum flow sampling for failure probability), arXiv:2605.16429, 2026.

[17] (Quantum amplitude estimation speedup for MPPI control), arXiv:2607.28851, 2026.

[18] M. Villén-Altamirano and J. Villén-Altamirano, “RESTART: A method for accelerating rare event simulations,” in Proc. ITC-13, 1991.

[19] M. C. Campi and S. Garatti, “The exact feasibility of randomized solutions of uncertain convex programs,” SIAM J. Optim. 19(3), 2008.

[20] H.-Y. Huang, R. Kueng, and J. Preskill, “Predicting many properties of a quantum system from very few measurements,” Nature Physics 16:1050–1057, 2020.

[21] H. Caesar et al., “nuScenes: A Multimodal Dataset for Autonomous Driving,” CVPR, 2020.

[22] Open X-Embodiment Collaboration, “Open X-Embodiment: Robotic Learning Datasets and RT-X Models,” arXiv:2310.08864, 2023.

[23] J. Meyer et al., “Quantum Natural Policy Gradients,” arXiv:2304.13571, 2023.